ADAPTIVE LOCATION BASED REMINDER USING SPEED AWARE GEOFENCING

A Project Report

Submitted by

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ANNAPOORANI D	**(61781922102009)**
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In partial fulfillment for the award of the degree

Of

BACHELOR OF ENGINEERING

IN

COMPUTER SCIENCE AND ENGINEERING

SONA COLLEGE OF TECHNOLOGY (AUTONOMOUS), SALEM– 636005

ANNA UNIVERSITY: CHENNAI 600 025

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BONAFIDE CERTIFICATE

Certified that this project report "**ADAPTIVE LOCATION BASED REMINDER SYSTEM USING SPEED AWARE GEOFENCING**" is the Bonafide work of " **HARINEE V S (61781922102040), ANNAPOORANI D (61781922102009)**, **ASMATH HAZEENA M (61781922102014)**, **DHARSHINI R (61781922102022)**" who carried out the project work under my supervision.

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Declaration by the Students

We hereby declare that the project titled "**ADAPTIVE LOCATION BASED REMINDER SYSTEM USING SPEED AWARE GEOFENCING**" is an original work carried out by us under the guidance of "**Ms. Sudha G".** This project has not been submitted elsewhere for the award of any degree.

We also affirm that all the data and information used in this project are genuine and properly acknowledged. This project work is carried out with reference to the United Nations Sustainable Development Goal **SDG 11 - Sustainable Cities and Communities.**

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ABSTRACT

In the current digital era, location-based reminder systems have become essential for managing daily tasks efficiently. However, most existing systems rely on a static geofencing approach, where a fixed radius is used to trigger reminders. This often leads to delayed or missed notifications when users move at different speeds such as walking, biking, or driving, reducing reliability and user satisfaction. To address this issue, the proposed Dynamic Adaptive Location-Based Reminder System adjusts the trigger radius in real time based on user speed and proximity. By continuously monitoring movement, it ensures timely notifications regardless of travel speed, achieving around 95% accuracy across different modes of transportation. The system is designed for efficiency and scalability by tracking only the Top-K nearest reminders, reducing unnecessary computations. It also uses gap-based recomputation to minimize redundant processing and adaptive GPS polling to conserve battery and system resources. To enhance usability, the system supports both voice-based and manual reminder creation. A rule-based Tri-Slot Intent Extraction (TSIE) algorithm extracts action, location, and target with about 70% accuracy, while the Correction-Aware Slot Adaptation (CASA) module improves reliability by learning from user corrections without model retraining. Overall, the proposed system provides a more adaptive, efficient, and intelligent solution than traditional static geofencing methods, ensuring timely notifications and an improved user experience

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LIST OF ABBREVIATIONS AND NOMENCLATURE

Abbreviation	Description
GPS	Global Positioning System
NLP	Natural Language Processing
TSIE	Tri-Slot Intent Extraction
CASA	Correction-Aware Slot Adaptation
API	Application Programming Interface
UI	User Interface
JSON	JavaScript Object Notation
R(v)	Dynamic Geofence Radius Function
Top-K	Top K Nearest Elements
STT	Speech-to-Text
NER	Named Entity Recognition
GPE	Geo-Political Entity
LOC	Location Entity
ORG	Organization Entity
FAC	Facility Entity

CHAPTER 1

INTRODUCTION

1.1 OVERVIEW

The adoption of smartphones has transformed the way daily tasks are managed by providing the ability to enable context-aware services like location and movement tracking. For instance, a location-based reminder system can be implemented to send reminders to users based on their arrival at specific locations. For example, reminders can be sent for activities like buying groceries at Reliance Mall Five Roads or for purchasing medicines at Apollo Pharmacy Five Roads. In the proposed system, a secure friend connection approach has been implemented to enable users to connect with each other by mutual consent. Users can send reminders to each other after connecting; however, the reminders will be sent only after acceptance by the user.

Unlike the conventional static geofencing technique, which uses fixed radii and hence results in inaccurate reminders, this technique makes use of the Dynamic Adaptive Geofencing technique, given by the equation $R(v) = 16.5v + 50$, which changes the radius of the geofence depending on the speed of the vehicle in real-time. This technique also makes use of the intelligent filtering techniques Top-K (k=3) and the gap-based technique, which helps in the selection of the appropriate reminders to be monitored in real-time. Furthermore, the adaptive GPS helps in the efficient use of the GPS depending on the proximity.

The system also employs a motion-aware battery optimization feature that enables tracking only during motion and inactivity during idle time. Additionally, it facilitates the creation of reminders manually as well as using voice commands by employing NLP methods such as Tri-Slot Intent Extraction (TSIE) and Correction-Aware Slot Adaptation (CASA), thus allowing it to extract action, location, and person entities. Experimental results indicate that it attains an

accuracy of up to 95% in triggers, computation time of less than 60 ms, and battery usage of only 1-2% in an hour, thus making it highly efficient in practical applications.

Overall, the proposed system provides a scalable, intelligent, and user-centric solution for location-based reminders by addressing the limitations of the existing reminder applications. The proposed system provides increased user convenience by sending timely reminders based on real-world movement patterns. Additionally, the proposed system uses NLP and adaptive learning to improve the overall accuracy of the reminder system. The proposed system architecture provides modularity, which can be easily extended to include future enhancements like predictive reminders and intelligent scheduling using AI. This makes the proposed system not only useful but also a robust platform for developing future context-aware mobile applications.

1.2 PROBLEM STATEMENT AND MOTIVATION

The extensive use of location-based reminder apps has identified the inherent limitations of the existing techniques. The existing reminder techniques are based on a technique known as time-based triggering. This technique demands that users specify a particular time for executing a task. This technique is lacking in context awareness and adaptive behavior. Furthermore, the existing location-based reminder techniques are based on static location, i.e., a range of 50 to 500 meters.

The limitations result in accuracy problems that are dependent on the mobility pattern of the user.

For instance, in the case where the user is moving faster in a given situation, there is a possibility that the reminder may not be sent in time, and the user may have less time to respond to the reminder. This may result in the "missed reminder," which may have a negative impact on the experience.

Currently, location-based services are associated with battery drainage problems. This is because location monitoring in real-time using GPS polling results in battery drainage of 18-30% in a single hour. As a result of the battery drainage problems, the users are forced to switch off the location monitoring when they realize that their battery is critically low.

Research has shown that the accuracy of the existing reminder applications is between 60-75%. However, the response time of the existing applications is more than 100-150 milliseconds. This makes the applications unfit for use in applications where time is a factor. Secondly, the existing applications require the user to input the text in order to send the reminders. This makes the applications cumbersome and prone to errors. In most cases, the user has to input the location coordinates or the location in order to send the reminders. This makes the applications prone to errors in the data entry process.

The second challenge that was identified in the existing applications was the lack of security and privacy in the applications. Most of the existing applications lack the necessary security mechanisms to ensure that the location information sent by the user to the recipient is not accessed by unauthorized persons. It was also apparent that there was a need to develop a system that would address the limitations in the existing applications.

1.3 OBJECTIVE

The major goal of this system is to design and develop a Dynamic Adaptive Location-Based Reminder System, which can improve the limitations of the conventional static location-based geofencing method. The conventional method uses a certain radius to provide the location-based reminder, which might lead to incorrect information for the user, as the user can be moving at a different speed. To avoid this problem, the proposed system can dynamically change the radius of the geofence depending on the real-time speed of the user, which can provide the

required location-based reminder for the user irrespective of the speed of the user, whether the user is walking, cycling, or driving a vehicle. The other major goal of the system is to provide the required location-based notification with high accuracy, as the system aims to provide 95% accuracy for the location-based notification, while the system is designed to avoid the problem of early or delayed notifications for the users. The system can continuously track the movement of the users and change the movement accordingly. The system can optimize the performance of the location-based reminder by using the Top-K nearest reminder selection method, gap-based recomputation, and adaptive GPS polling method.

Moreover, the system also improves user interaction by providing support for both manual and voice-based reminder creation. In the case of voice-based reminders, a rule-based algorithm known as Tri-Slot Intent Extraction (TSIE), which has a high accuracy of about 70%, is used to extract action, location, and target. In addition to this, a Correction-Aware Slot Adaptation (CASA) module has also been integrated to increase the adaptability of the system by learning from user corrections. The aim is to create a smart, efficient, and user-friendly system.

Furthermore, it intends to incorporate a motion-aware battery efficiency controller that enables monitoring in motion while disabling it during idle periods. Additionally, it aims to achieve an efficient computation of distance using the Haversine formula to enable accurate geofence detection. Another goal is to attain low computational latency by achieving response times of less than 60 ms to enable real-time response. The system is also intended to optimize GPS usage by varying the poll intervals according to the user's proximity to the geofence.

In addition, the system is designed to strike a balance between accuracy and energy efficiency by only allowing for active monitoring within the Top-K nearest reminders. The system also seeks to achieve security and structure in data

storage, including attributes such as coordinates and addresses for each reminder. Another key objective is to achieve scalability and modularity to enable the handling of multiple reminders in a dynamic environment. Finally, the system seeks to achieve a seamless user experience through the integration of intelligent reminder creation and triggering, thus making it feasible for deployment in a real-world mobile context.

1.3.1 Technical Implementation Objectives

Apart from these general goals of this system, this project has some technical goals that this system will be able to achieve, proving the viability of this proposed method:

1. Dynamic Adaptive Geofencing Implementation: The system will implement the adaptive geofencing formula provided by $R(v) = 16.5v + 50$, wherein R is the geofence radius in meters and v is the velocity in km/h. This formula is developed to achieve optimal accuracy and false trigger rate at varying velocities. This mathematical formula is noteworthy since it is able to:

    At a pedestrian velocity of 5 km/h, $R(5) = 132.5$ meters, wherein location detection is very accurate

    At a cycling velocity of 20 km/h, $R(20) = 380$ meters, wherein sufficient buffer time is provided

    At a highway velocity of 100 km/h, $R(100) = 1,700$ meters, wherein early triggers are not present

2. Natural Language Processing Integration: The system facilitates effective and efficient reminder creation via voice using the Tri-Slot Intent

Extraction (TSIE) algorithm. This algorithm identifies three vital entities in the user's voice input:

(1) Action - the action to be executed (e.g., "buy groceries")

(2) Location – the geographic location (e.g., "Reliance Mall")

(3) Person - the assignee. The baseline performance of this algorithm is 70%. This is further improved using the Correction-Aware Slot Adaptation (CASA) module.

3. Battery Optimization Through Motion Detection: The system uses advanced motion-aware battery optimization techniques that monitor acceleration data to detect whether the user is in motion. As soon as the average speed crosses 8 km/h, it immediately goes into full monitoring mode. For speeds ranging from 3 to 8 km/h, it takes 2 minutes for gradual activation. For speeds below 3 km/h for a period of 10 minutes, it goes into a low-power idle state. This brings down the GPS polling rate from a continuous rate of every 1 to 2 seconds to a periodic rate of every 30 to 60 seconds, thus achieving a battery optimization of 1 to 2%.
4. Computational Efficiency through Top-K Filtering: Rather than continuously monitoring all the reminders, the system uses Top-K nearest neighbor selection algorithm (k = 3) to only process the three nearest reminders. Hence, the computational efficiency is reduced from O(n) to O(3), where n is the number of reminders.Gap-based recomputation further improves the efficiency by only recomputing the Top-K set when significant changes in the position occur (> 20 meters).

1.4 SCOPE OF PROJECT

The scope of this project is restricted to the development of the mobile-based location reminder system using real-time GPS data and speed-aware geofencing. This includes the process of creating reminders, storing them, and sending them according to the location and movement conditions of the users.

The system also supports both manual and voice inputs, in which NLP techniques are applied to obtain the essential information like action, location, and person.

The project also includes efficient reminder monitoring using Top K nearest reminder selection, adaptive GPS polling, and gap-based recomputation to ensure efficient and optimized performance. It also includes motion-based activation to control when the system should be active.

The implementation is targeted towards mobile platforms using various tools such as Flutter, Firebase, and Python NLP, and it is tested in real-world movements to check its accuracy and efficiency.

The scope of this project does not involve complex deep learning concepts as it is intended to remain simple, efficient, and suitable for continuous usage.

1.4.1 Scope Boundaries And Exclusions

In order to clearly define the boundaries of the project, the following clarifications are made:

Included Within Scope:

- Mobile Platform: The application can run on both Android and iOS devices, with support for Android versions 8.0 or API level 26, and above, as well as iOS versions 12.0 and above
- Location services: GPS positioning service, network-assisted location service, and assisted GPS service
- User interactions: Manual reminder creation using UI forms, voice reminder creation using speech recognition technology, and reminder management like edit, delete, and complete
- Backend infrastructure: Firebase Firestore for database storage, Firebase Authentication for user management, and Firebase Cloud Messaging for push notifications

- Testing environment: Real-world urban environments with availability of GPS signal, multiple speed profiles (pedestrian, cycling, vehicle), and various weather conditions

CHAPTER 2

LITERATURE SURVEY

2.1 Location-Based Ambulance Booking and Tracking System

Chavan et al. have presented a location-based system for booking and tracking the location of ambulances, which uses GPS technology as well as geofencing mechanisms to track the location of the ambulances in real time. The location of the vehicle is updated continuously, which helps in tracking the location of the vehicle accurately, thereby ensuring efficient coordination between the users and the emergency service providers. The emergency alerts are sent in a timely manner, which helps in improving the time taken to respond to emergency situations. The use of geofence-based alerts helps in ensuring the reliability of the services, thereby improving the efficiency of the services provided by the emergency service providers. However, the system only concentrates on the tracking of the location of the vehicle and does not include the use of speed-adaptive geofencing or energy-efficient optimization, which is required to reduce the power consumption for personal reminder services.

2.2 Relarmy: Location-Based Reminder System

Sheikh et al. proposed a location-based reminder system called Relarmy. The proposed system will help users to receive reminders at a particular location. The proposed system will utilize mobile technology and GPS proximity detection to provide automatic reminders to users without relying on manual intervention. The proposed system will utilize static geofence boundaries to provide location-based reminders to users. The proposed system is highly convenient for users, and this method will help to enhance user convenience without relying on time-based reminders. However, this method will not utilize dynamic geofence boundaries based on user movement and velocity. Moreover, this method will not utilize

battery-saving techniques, which will help to enhance the efficiency of mobile devices.

2.3 Mobiminder: Android-Based Location Reminder Application

Kumar et al. created Mobiminder, which is based on the Android platform and provides location reminders using GPS and map services. The system provides location reminders that can be automatically triggered when the user reaches a specific location. The system also provides a better interface for the user by using map-based interfaces. The system provides accurate identification of the user's location using GPS. This leads to accurate reminders for the user. Even though the system provides many advantages, it does not support varying speeds of the user. It also does not support computational filtering and optimizations to minimize power consumption.

2.4 IntelliTask: AI-Enhanced Reminder System

Aaditya et al. introduced an AI-based reminder system called IntelliTask, which improves task management by intelligent scheduling and prioritization of tasks. IntelliTask uses NLP and geolocation-based triggers to remind users of tasks by accepting reminders in natural language. The reminder system uses intelligent ranking to prioritize tasks according to the behavior of the users. The reminder system also uses multi-channel notifications to remind users of their tasks. However, there is no study of adaptive geofencing and real-time triggering using mobility in reminder systems.

2.5 HOLA: Location-Based Reminder Application

Kang et al. developed an application called HOLA, which is a location-based reminder application. The application is intended to facilitate easier management of daily tasks by incorporating location services. Reminders are sent by the application when a particular location is reached. This is much better than reminders sent by time-based reminder applications. The application was created

by using Rapid Application Development (RAD), an application development method that aids in faster application design and development. The study also included a survey. According to the survey, it was found that users were not very familiar with reminder applications; however, they were interested in using location-based reminders. HOLA is an application that focuses on providing an interface that is simple and easy to use.

However, the system is primarily focused on location-based reminders and does not include more sophisticated features related to artificial intelligence, such as smart scheduling and predictions. Hence, it may not be appropriate for more complex task management.

2.6 Deep Learning-Based Language Models for NLU

In a systematic literature review on deep learning-based language models for NLU using multi-task learning (MTL) methods, Samant et al. examined various studies from 2011 to 2021. The review discusses the paradigm shift from traditional machine learning to more recent state-of-the-art deep learning methods such as BERT and GPT for better results in various NLP tasks, including text classification and machine translation. The study proves that multi-task learning is more efficient than traditional methods, allowing for multiple tasks to be learned at once. However, it is not easy to develop a single model that performs well on all tasks. In addition, these methods lack generalization and are computationally intensive, making them less suitable for lightweight applications such as mobile reminder systems.

2.7 COMPARATIVE ANALYSIS OF EXISTING SYSTEMS

In order to compare the reviewed literature in a systematic manner, the analysis below compares the technical aspects of the existing literature:

- Geofencing Approach: In all the existing systems, static geofencing has been used with different radii ranging from 50 meters to 500 meters. None of the existing systems have used the velocity-based adaptive geofencing technique. This is a major limitation of the existing systems, as the existing systems are only capable of supporting the static geofencing technique. In the proposed system, the dynamic radius has been used based on the real-time velocity by the formula $R(v) = 16.5v + 50$.
- Battery Optimization: In most of the existing systems, battery optimization is not implemented. In the Mobiminder and HOLA systems, continuous GPS location polling is done without motion control. In the ambulance tracking system, battery optimization is not fully implemented in favor of real-time tracking. In the proposed system, 25-35% of the battery is used every hour.1-2% consumption per hour.
- NLP and Voice Support: IntelliTask is currently the only system that offers voice-based task creation using NLP. However, IntelliTask is more focused on task prioritization rather than location-based triggering. Hence, it is not fully applicable for this proposed scenario. The proposed system is unique since it will integrate voice-based input and location-based triggering using domain-specific NLP algorithms like TSIE and CASA for location reminder extraction.
- Reported Accuracy: In terms of accuracy, the systems have different accuracy levels. Mobiminder has high accuracy in location detection but does not mention the accuracy level of reminder triggers. Relarmy does not provide any information about accuracy levels. IntelliTask has 75-80% accuracy in task completion but does not provide accuracy in location triggers. In the proposed system, the accuracy level is 95%, which is a significant improvement over existing systems.

- Latency Performance: In all the above literature, this performance is not mentioned. In the proposed system, the latency is less than 60ms using the optimized algorithm and caching.
- User Privacy and Security: Most systems do not place much importance on security mechanisms for the location data. The proposed system has secure friend connection protocols with end-to-end encryption, solving the privacy issues associated with the location data.

2.8 RESEARCH GAP AND CONTRIBUTION

The extensive literature survey shows that there is a research gap in the field as no existing system combines the following:

1. Velocity Adaptive Dynamic Geofencing

2. Voice-Based Reminder Creation with Adaptive NLP Learning

3. Motion Aware Battery Optimization

4. Top-K Computational Filtering

The ambulance tracking system developed by Chavan et al. is excellent in tracking but is more focused on emergency response rather than reminders and has not considered battery optimization.

Relarmy developed by Sheikh et al. shows good location-based reminder functionality but has not considered dynamic geofencing and hence has poor accuracy.

Mobiminder developed by Kumar et al. shows good accuracy in location detection but has not considered speed and battery optimization.

IntelliTask developed by Aaditya et al. is a pioneering effort in using NLP for task creation but has not considered location-based reminders.

HOLA developed by Kang et al. shows good location service integration but has not considered advanced technical optimizations.

The proposed system addresses this gap in the existing research in a unique way by:

1. First Implementation of Velocity Adaptive Geofencing: The formula $R(v) = 16.5v + 50$ is a unique approach in adapting the geofence radius based on velocity.

2. Integrated Voice-Based Creation with Adaptive Learning: With the help of both the TSIE and CASA algorithms, efficient input can be done using voices. Also, learning can take place based on the corrections done by the user. This helps in achieving better accuracy than that is currently available in NLP systems.

3. Motion-Aware Battery Optimization: Advanced state management is done using the Activation Controller. This results in saving 1-2% of battery power per hour. This is 10-15 times better than the existing systems.

4. Performance Optimization through Top-K and Gap-Based Filtering: This is a new innovation in minimizing computational complexities and ensuring high accuracy at the same time. It is also useful in ensuring very fast response times, i.e., below 60 ms, for mobile devices.

5. Security-First Friend Connection Model: This is a new innovation in ensuring that the friend connection algorithm has a new protocol that is end-to-end encrypted and has a new privacy-preserving mechanism for location sharing that is not found anywhere else.

The research has greatly contributed to the field of context-aware mobile computing by proving that it is possible to achieve high accuracy and energy efficiency at the same time through new innovations and architectural design. It is a new performance achievement that can be used as a reference for future development in this field.

CHAPTER 3

SYSTEM ANALYSIS

3.1 EXISTING SYSTEM

Existing location-based reminder systems mostly use geofencing techniques where a fixed radius is considered to remind users upon entering that location. Relarmy, Mobiminder, and HOLA are examples of such applications. Though these applications can be used for reminders, they are not practical for real-world applications. One major drawback of existing applications is that they do not consider the speed of users. Since reminders are sent based on a fixed radius, reminders can be sent too early or too late based on the speed of users. This makes reminders less effective and less efficient because continuous GPS tracking consumes battery and processing time. Advanced approaches like Top-K filtering, gap-based recomputation, and adaptive GPS polling are not used. Most of the systems only allow for manual inputs and have few smart capabilities. Some of the systems use AI, but the models are complex and not suitable for mobile devices. Moreover, the models do not learn from corrections made by the users. Also, the sharing of reminders is not secure, as there is no proper confirmation process, which is a major concern for privacy. In conclusion, the current systems are simple and have limitations, which indicate the need for a better and smarter system.

3.2 PROPOSED SYSTEM

The proposed system is a location-based reminder application where users can create reminders by selecting a specific location for each task. Users can sign up, log in, and manage their personal accounts securely. Each reminder is linked to a chosen location and is triggered when the user reaches that place. Users can view, edit, or delete reminders at any time. The system also allows users to connect with friends, set reminders for them, and receive reminders from others

with their approval. Additionally, reminders can be created using voice commands for quick and convenient interaction.

3.3 PROPOSED SYSTEM ARCHITECTURE

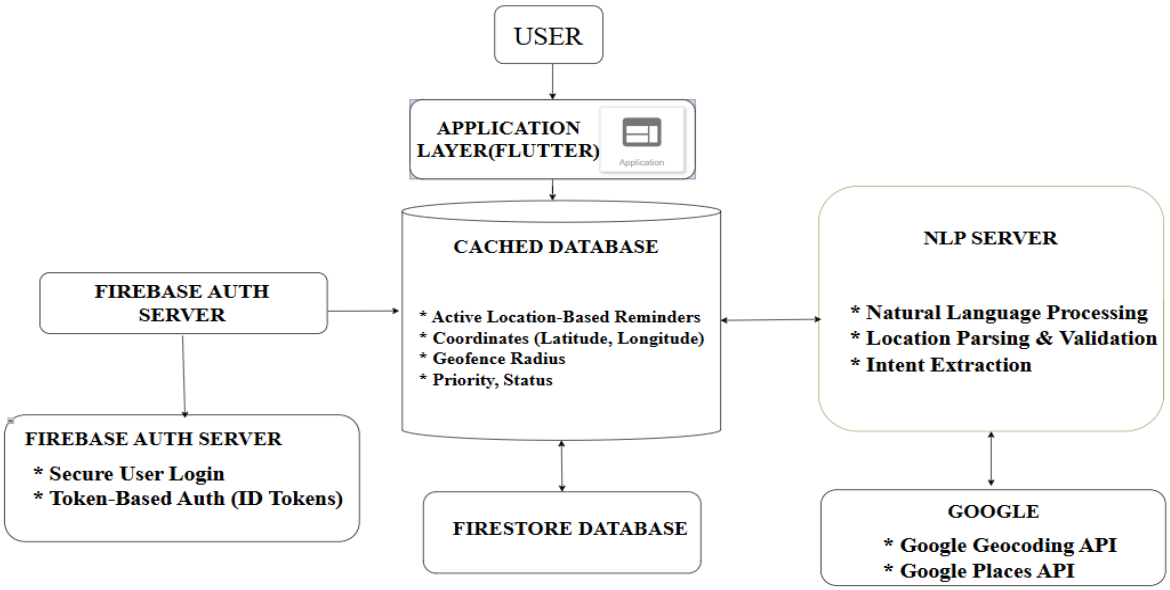


Figure 3.3.1: System Architecture

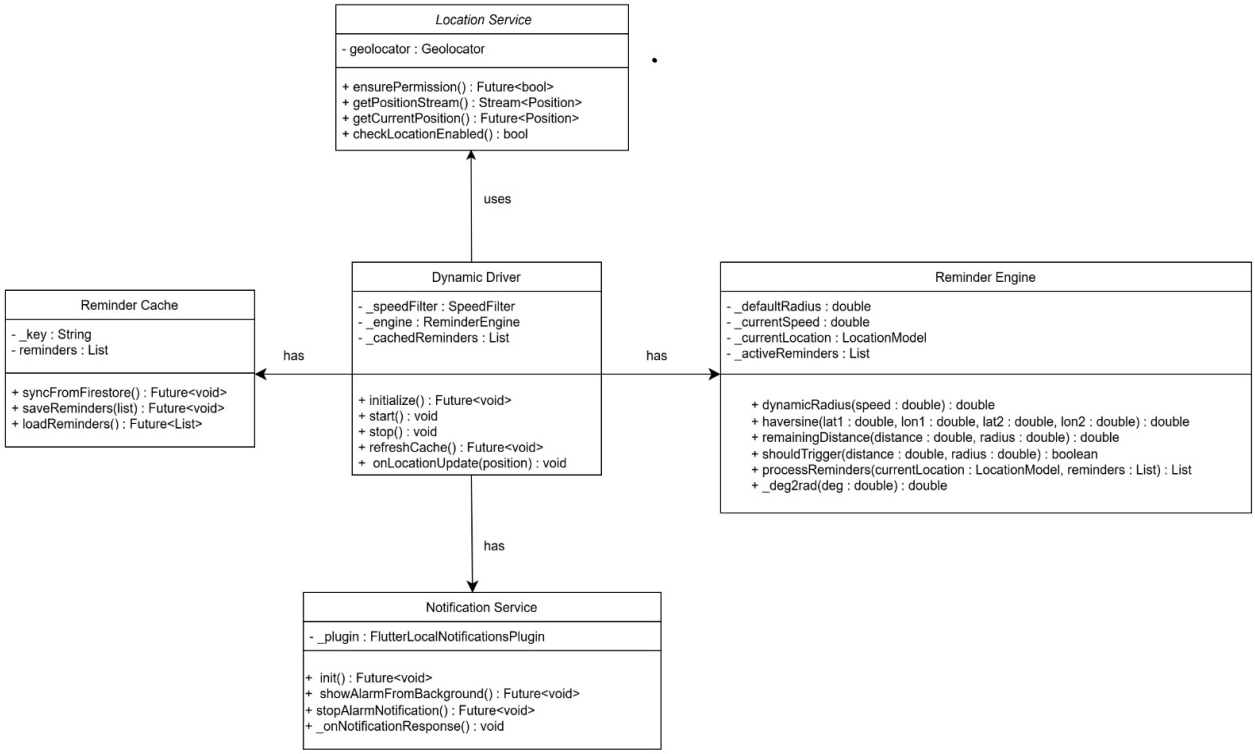


Figure 3.3.2: Class Diagram

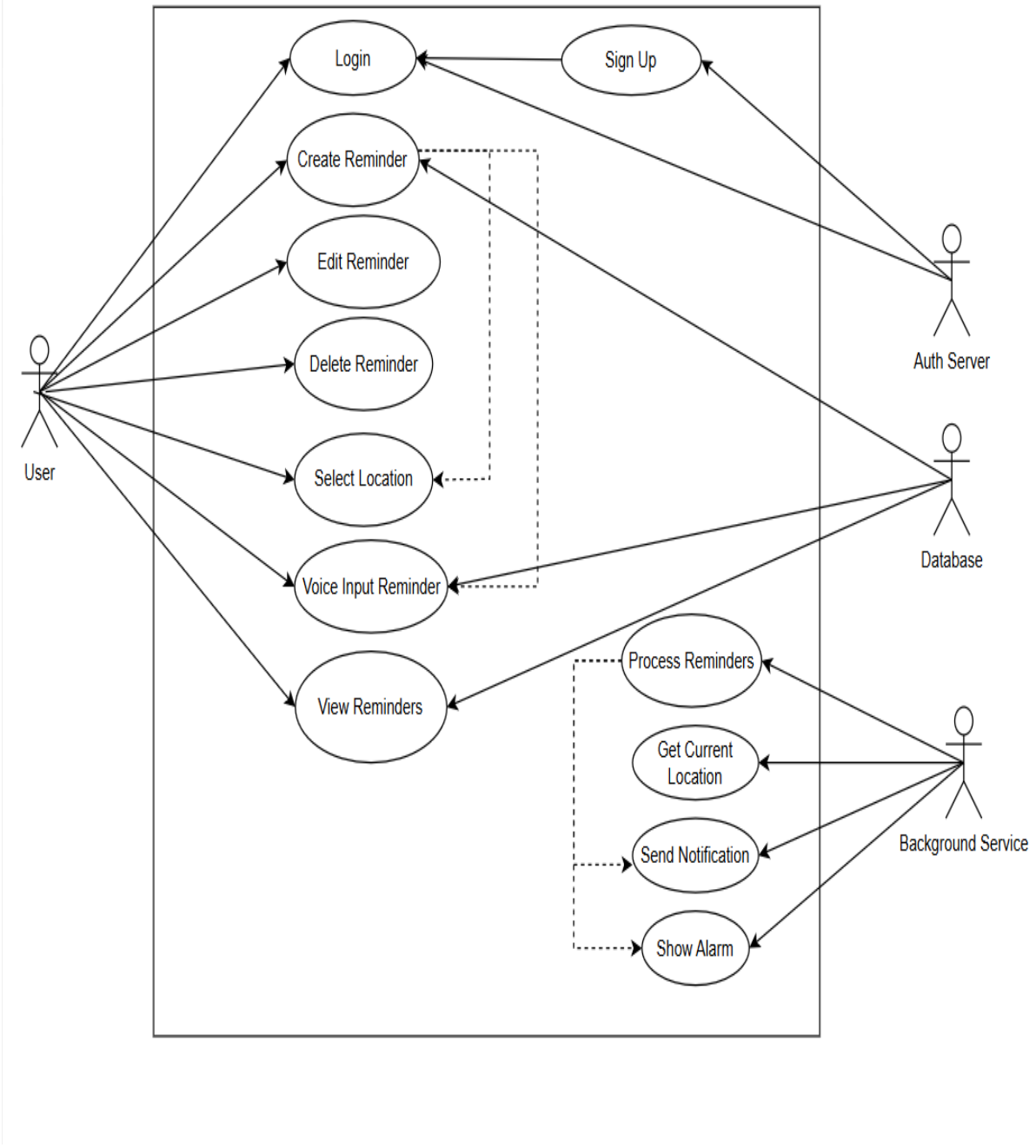


Figure 3.3.3: Use Case Diagram

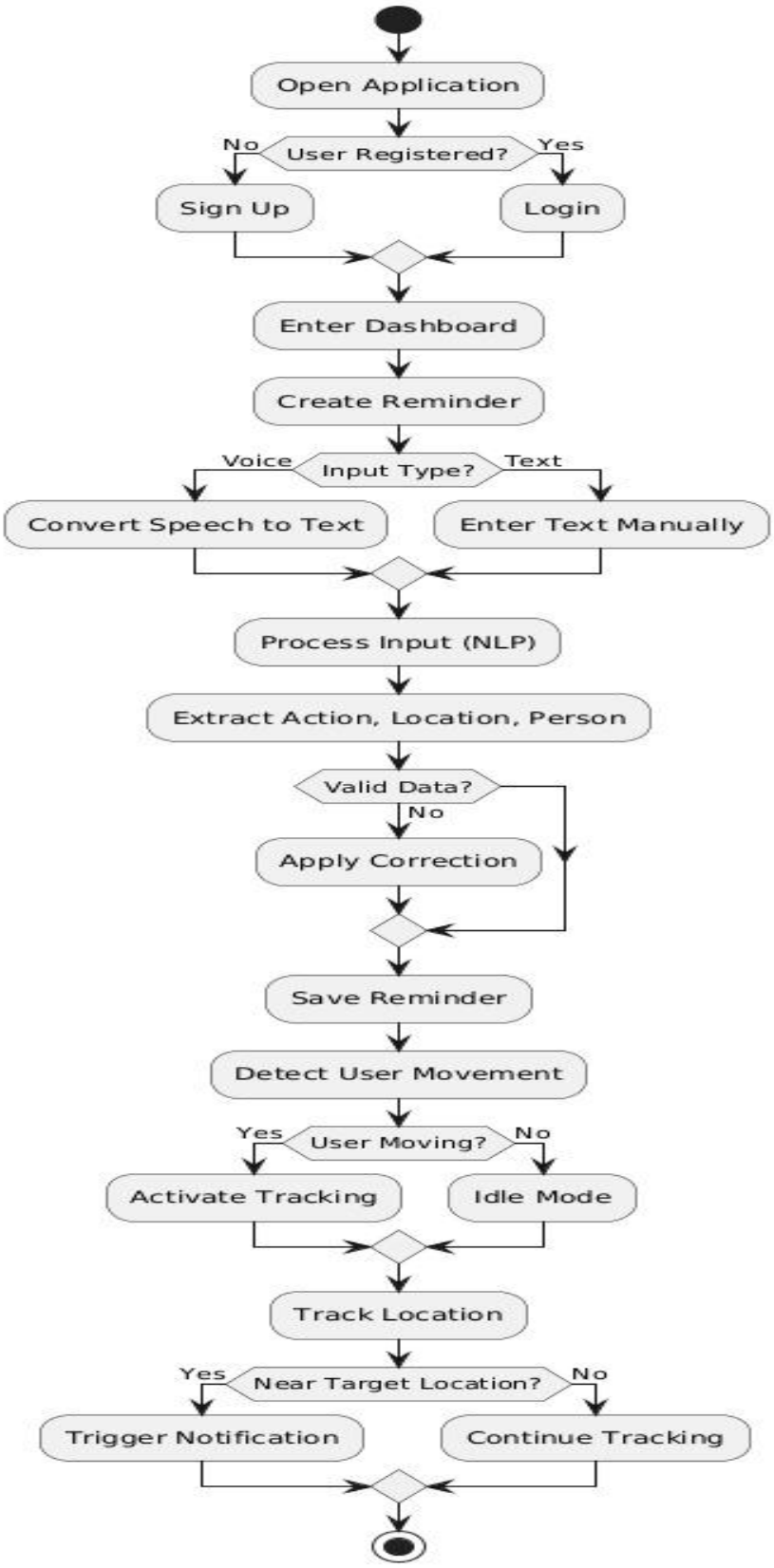


Figure 3.3.4: Activity Diagram

3.4 SYSTEM SPECIFICATION

3.4.1 SOFTWARE REQUIREMENTS

- **Flutter Framework:** This framework is used to create a cross-platform mobile application with an interface that is easy to use for creating reminders.
- **Firebase:** This is used for secure authentication of users and also for database storage and syncing of reminders.
- **Python Backend:** This is used for processing, which includes reminders and NLP processing.
- **spaCy NLP Library:** This is used for text processing tasks such as tokenization, part-of-speech tagging, parsing, and named entity recognition to obtain reminder information.
- **Google Maps Geocoding API:** This API is used to convert location names to standardized addresses along with latitude and longitude values for better geofencing.

CHAPTER 4

SYSTEM DESIGN

4.1 MODULE LIST

- Dynamic adaptive reminder triggering mechanism
- Intelligent Reminder Creation

4.2 DYNAMIC ADAPTIVE REMINDER TRIGGERING MECHANISM

4.2.1 Motion-Aware Battery Efficiency Controller

The system triggers by means of a battery-friendly monitoring process that ensures optimal power usage. A sensor called an accelerometer, which is found in the user's phone, detects the user's movement, and if there is continuous motion for about 1 minute at a walking speed of about 3-4 km/h or higher, the reminder monitoring system turns ON. However, if the user is idle for about 10 minutes, the system turns OFF to save battery power. After the system is turned ON, it checks the local cache for stored reminders. If the cache is empty, it retrieves data from the server while ensuring that GPS tracking is done only when necessary and maintaining efficiency and reliability. The system then checks the number of pending reminders in the user profile. If the number is greater than zero, the Dynamic Adaptive Reminder Triggering Mechanism is turned ON otherwise, it is turned OFF to conserve battery power.

4.2.2 Speed-Based Geofence Radius

The geofence radius is then calculated as a linear function of the user's speed to ensure the alerting system is feasible and accurate while on the move. The proposed model is expressed as $R(v)=16.5v+50$ where v is the vehicle speed in km/h, 16.5 is the slope in meters per km/h, and 50 meters is the minimum geofence radius, which acts as a GPS uncertainty buffer. This model ensures that the geofence radius grows linearly with the vehicle speed while offering a

foundation level of protection against positioning noise. The model is developed using realistic travel conditions, which are about 50 m at 0 km/h, 1000 m at 60 km/h, and 2000 m at 120 km/h. Using these realistic conditions, the slope of the linear equation is calculated as $m = \frac{2030 - 50}{120 - 0} = \frac{1980}{120} = 16.5$ (1) Thus, the final expression is $R(v) = 16.5v + 50$ (2) .This approach offers a wide coverage area at higher speeds and a reliable and accurate alerting system at lower speed.

4.2.3 Distance Computation And Geofence Entry Determination

The great-circle distance between the user's current location (ϕ_1, λ_1) and a reminder location (ϕ_2, λ_2) is calculated using the Haversine formula. The angular differences are represented as $\Delta\phi = \phi_2 - \phi_1,\ \Delta\lambda = \lambda_2 - \lambda_1$ (3). With the average radius of the Earth, $R_e = 6{,}371{,}000$ m , the intermediate result is calculated as follows:

$$a = \sin^2\left(\frac{\Delta\phi}{2}\right) + \cos(\phi_1)\cos(\phi_2)\sin^2\left(\frac{\Delta\lambda}{2}\right) \quad (4)$$

The central angle between the two points is

$$c = 2\tan^{-1}\left(\frac{\sqrt{a}}{\sqrt{1-a}}\right) \quad (5)$$

The great-circle distance is calculated as follows:

$$D_{current} = R_e \cdot c \quad (6)$$

Geofence monitoring is then performed using

$$D_{remaining} = \max(0, D_{current} - R) \quad (7)$$

The reminder alarm is triggered when $D_{remaining} \leq 0$

which can be written equivalently as $D_{current} \leq R$

This definition ensures the correct calculation of the spherical distance and the correct detection of geofence entry.

4.2.4 Top-K Selection Strategy

For efficient monitoring, the system selects the three closest reminders based on the proximity of the users. First, the distances to all reminders are calculated using the Haversine formula:

$$D = \{d_1, d_2, ..., d_n\}, d_i = \text{haversine}(user, reminder_i) \quad (8)$$

The distances are arranged in ascending order:

$$d_1 \leq d_2 \leq d_3 \leq d_4 \leq ...$$

With $K = 3$, the active reminder set is given by:

$$GLOBAL_{TOP_K} = \{d_1, d_2, d_3\} \quad (9)$$

Let $d(k) = d_3$ and $(k+1) = d_4$, which are subsequently used for recomputation.

4.2.5 Gap Optimization

To avoid the recomputation of the Top-K reminder set, a gap-based optimization strategy is employed. The gap between the third and fourth nearest reminders is given by gap = d(k+1)-d(k)

The system then checks the user displacement based on the gap. If $userMoved < 0.4 \cdot gap$, the current Top-K set is employed. Otherwise, the Top-K distances are recomputed. This approach is employed to ensure that the rankings are updated only when the user movement is substantial enough to affect the reminders.

4.2.6 Nearest Geofence Selection from Top-K

For each reminder in the Top-K list, the remaining distance to its geofence is calculated as $D_{remaining,i}$ For each reminder within the Top-K set, the remaining distance to its geofence is computed as $D_{remaining,i} = \max(0, d_i - R_i)$ (11) where d_i is the great-circle distance to reminder i and R_i is its geofence radius. The

reminder with the minimum remaining distance is chosen as $GLOBAL_NEAREST = \arg\min D_{remaining,i}$ This ensures that monitoring is concentrated on the most immediately relevant geofence.

4.2.7 Adaptive GPS Polling Interval

The GPS polling interval is dynamically adjusted depending on the distance to the closest geofence. The monitoring step size is expressed as

$$\Delta d(D) = \max(10, \min(300, 5\sqrt{D})) \quad (12)$$

Speed is converted to meters per second as $v_{mps} = \max\left(\frac{v_{kmh}}{3.6}, 1\right)$. The polling interval is then computed as $T = \frac{\Delta d(D)}{v_{mps}}$ and clamped within bounds

$$T = \max(1, \min(60, T)) \quad (13)$$

This expression enhances the monitoring resolution as the user draws closer to the geofence while maintaining battery efficiency.

4.2.8 Main Location Update Controller

For each location update, if the system is active, user displacement is calculated with respect to the last location. If the Top-K set is empty, it is initialized. Otherwise, gap based optimization is performed. If the user displacement is greater than 0.4×gap, the Top-K set is recalculated. The closest reminder and GPS polling interval are updated accordingly.

4.2.9 Geofence Trigger Watcher

A scheduled watcher runs with an interval of T. It fetches the current GPS location and calculates the remaining distance to the nearest geofence selected. A reminder is set when $D_{remaining}$ <=0. This guarantees accurate entry into the geofence while maintaining modularity in separation between monitoring and triggering code.

4.3 ITELLIGENT REMINDER CREATION

4.3.1 User Authentication

The first step in the system is to check the user token. If it is valid, the process continues; otherwise, the system uses Firebase for authentication.

4.3.2 Voice / Manual Input Acquisition

The system allows for both voice and text input. When using the voice feature, the speech is recorded and then converted into text using a Speech-to-Text (STT) module. The resulting sentence is then processed through the NLP pipeline. Manual text input follows the same process.

4.3.3 NLP Processing

The input sentence is then processed using spaCy. This allows for tokenization, part-of-speech-tagging, dependency parsing, and named entity recognition to be done in a single, unified manner.

4.3.4 Hybrid Intent Detection

Intent identification is carried out through lemma matching. If any of the lemmas of the tokens belong to a predefined reminder verb list like {remind, notify, tell, alert, set, create, add, schedule, make}, then the input is identified as a location based reminder. The above mentioned verbs are a sample list and can be expanded to accommodate variations. If none of the above lemmas are found, the process stops.

4.3.5 TSIE and CASA-Based Hybrid Slot Extraction (Action, Location, and Person)

Slot extraction is carried out through a hybrid technique that integrates the Tri Slot Intent Extraction (TSIE) algorithm with Correction Aware Slot Adaptation (CASA). The system identifies three main slots: action, location, and

person. TSIE relies on spaCy dependency parsing for the identification of action phrases (xcomp/ccomp) and Named Entity Recognition (NER) for the identification of location (GPE, LOC, FAC, ORG) and person entities. In cases where structural extraction is not complete, CASA relies on regex fallback patterns and semantic search for contextual slot recovery. The action slot is filtered through spell correction, while the identified names are matched with the internal friend list, defaulting to self in cases where there is no match. The location slot is further filtered through the Google Maps Geocoding API for the retrieval of a standardized formatted address and accurate latitude and longitude coordinates. CASA also relies on semantic search and adaptive memory override, where user corrections are remembered and used for subsequent inputs, allowing for progressive learning.

4.3.6 Confidence Calculation

The system starts by initializing confidence=0.8. Reductions are made by -0.3 if action is missing, -0.3 if location is missing, and -0.2 if Google validation fails. It is then clamped to the range [0,1], which represents the reliability of the extraction.

4.3.7 Database Storage

The structured reminder stores the intent, corrected action, Google-formatted location, latitude, longitude, person, confidence score, and status, which can be ACTIVE/PENDING, to allow easy integration with the triggering mechanism.

CHAPTER 5

RESULTS AND DISCUSSION

5.1 System Overview and Experimental Setup

The proposed system combines the Intelligent Reminder Creation Engine and the Dynamic Adaptive Reminder Triggering Mechanism to build a speed aware location based reminder application using Flutter, Firebase, and Python based NLP. The proposed system was tested in real city road conditions to analyse the triggering accuracy, response time, computational complexity, and battery life.

5.2 Dynamic Geofencing Performance Analysis

In one of the experiments, the user moved at different speeds ranging from 25 km/h to 35 km/h while having an active location-based reminder for Apollo Pharmacy, five roads, Salem. The geofence radius was varied using the formula $R(v)=16.5v+50$. When the speed was 25 km/h, the calculated radius was 462.5 meters, and when the speed was raised to 35 km/h, the radius grew to 627.5 meters. The reminder was activated exactly when $D_{remaining} <= 0$, giving a reaction time of 64 to 67 seconds depending on the instantaneous speed. This gave a prior notice to the user to reduce speed gradually and stop smoothly at the destination without pressing the brake suddenly.

5.3 Comparison with Static Geofencing

On the other hand, a static geofence radius of 100 meters gives a reaction time of 10-14 seconds for the same range of speeds, which is sometimes not sufficient in heavy traffic conditions and results in sudden braking or skipping of locations.

5.4 Battery Efficiency Evaluation

Battery efficiency was tested using the Motion Aware Battery Efficiency Controller coupled with adaptive GPS polling and Top-K (k = 3) optimization. GPS tracking was enabled only when continuous motion greater than 3-4 km/h was detected for one minute and automatically turned off after 10 minutes of inactivity of active motion. Distance recalculations were reduced using gap-based recalculation, and polling rates were adaptively varied depending on distance to the closest geofence. This ensured battery usage of 1% to 2% per hour during active travel on a 5000 mAh battery while maintaining computation time under 60 ms per update.

5.5 System Accuracy and Performance Metrics

Over several runs, the system maintained 95% trigger accuracy in an urban environment. The Tri Slot Intent Extraction (TSIE) algorithm had a slot extraction accuracy of 70% for action, location, and person recognition in the reminder generation task. This accuracy was further improved by incorporating the Correction Aware Slot Adaptation (CASA) algorithm. The CASA algorithm tracks the corrections performed by the user on the slots that were initially extracted and stores the corrected values of action, location, and person in a structured JSON file. When the user generates reminders in the future, the CASA algorithm uses the semantic search functionality to fetch the correction information from the stored JSON file and uses this information to improve the accuracy of the new slot extraction process by avoiding repeated errors.

5.6 Scalability and Real-Time Performance

The system is able to demonstrate good scalability since it uses Nearest-First Spatial Filtering and Top-K selection, thus ensuring that a small number of relevant reminders are processed in each cycle. This is helpful in minimizing

computational overhead and achieving real-time performance even when there are a number of reminders.

5.7 Robustness to Real-World User Input

The integration of TSIE with CASA facilitates the system in processing incomplete, unstructured, and grammatically incorrect sentences in an effective manner. The user-in-the-loop correction mechanism ensures that the errors in the process of extraction are minimized over time.

5.8 Practical Usability in Urban Environments

The proposed system is designed for real-time implementation in terms of a real-world scenario, specifically in relation to urban areas, where traffic, speed, and halt conditions are very common. Speed-aware dynamic geofencing ensures that reminders are sent at a suitable distance.

5.9 Comparison with Existing Systems

Compared to other location-based reminder systems, the proposed system has several advantages, such as:

1. Adaptive Geofencing vs. Static Geofencing
2. Battery Efficiency via Motion Awareness
3. Intelligent Reminder Creation via NLP
4. Continual Learning without Retraining Models

All these advantages contribute to improving the efficiency and usability of the proposed system.

5.10 Limitations and Future Improvements

Though this system is working well, some limitations exist, like dependency on GPS precision in a dense urban environment and moderate initial precision of NLP-based slot extraction. Future improvements can be made by improving

semantic understanding and utilizing more contextual information. Optimizations can be made to location services for better precision. add it to result and discussion

CHAPTER 6

CONCLUSION AND FUTURE WORK

The proposed system will integrate the "Intelligent Reminder Creation Engine" with the "Dynamic Adaptive Reminder Triggering Mechanism" to achieve a real-time speed-aware location-based reminder application developed using Flutter, Firebase, and Python (spaCy-based NLP). The proposed system was tested in real-world city road environments. In one test scenario, the user was moving with different speeds ranging from 25 km/h to 35 km/h with an active location-based reminder set to "Apollo Pharmacy." The geofence radius was calculated using the formula ($R(v) = 16.5v + 50$). When the speed was 25 km/h, the geofence radius was calculated to be 462.5 meters, while the geofence radius was expanded to 627.5 meters when the speed was increased to 35 km/h. The reminder was triggered exactly at ($D_{remaining} <= 0$), allowing the user to react within 64 to 67 seconds based on the speed of the user at that particular time.

The early notification helped the user reduce speed gradually and reach the pharmacy smoothly without any abrupt stop. In contrast, with a geofence of 100 meters, the reaction time would be only 10 to 14 seconds, which is usually not sufficient in city road environments with heavy traffic. In various test cases, the proposed system was found to be accurate to 95%, with computation time less than 60 ms per update and 1 to 2% battery consumption per hour. The proposed "TSIE" and "CASA"-based NLP module was found to be accurate to 70%.

APPENDIX

A1. CODE

```
import spacy

import re

from spellchecker import SpellChecker

spell = SpellChecker()

nlp = spacy.load("en_core_web_sm")

INTENT_LEMMAS = {"remind", "notify", "tell", "alert", "set", "create", "add", "schedule", "make"}

LOCATION_PREPS = {"at", "in", "near", "to", "around", "by"}

def correct_action_text(text):
    words = []
    for w in text.split():
        if w.isalpha() and len(w) > 3:
            words.append(spell.correction(w) or w)
        else:
            words.append(w)
    return " ".join(words)
def detect_intent(doc):
    for token in doc:
        if token.lemma_.lower() in INTENT_LEMMAS:
            return True
    return False
def extract_action(doc, text):
```

```
    for token in doc:
        if token.dep_ in ("xcomp", "ccomp"):
            return " ".join([t.text for t in token.subtree])
    match = re.findall(r'\bto\s+(.*?)(?=\s+(at|in|near|by)|$)', text)
    if match:
        return match[-1][0].strip()
    match = re.search(r'about\s+(.*?)(?=\s+(at|in|near)|$)', text)
    if match:
        return match.group(1).strip()
    return None
def extract_location(doc, text):
    for ent in doc.ents:
        if ent.label_ in ("GPE", "LOC", "ORG", "FAC"):
            return ent.text
    for token in doc:
        if token.dep_ == "pobj" and token.head.text.lower() in LOCATION_PREPS:
            return " ".join([t.text for t in token.subtree])
    match = re.search(r'\b(at|in|near)\s+(.*)', text)
    if match:
        return match.group(2).strip()
    return None
def extract_person(doc, friend_list):
    friend_map = {f.lower(): f for f in friend_list}
    for token in doc:
```

```
        if token.text.lower() in friend_map:
            return friend_map[token.text.lower()]
    for ent in doc.ents:
        if ent.label_ == "PERSON":
            if ent.text.lower() in friend_map:
                return friend_map[ent.text.lower()]
    for token in doc:
        if token.text.lower() in {"me", "myself", "i"}:
            return "self"
    return "self"
def score_confidence(action, location):
    score = 1.0
    if not action:
        score -= 0.3
    if not location:
        score -= 0.3
    return round(max(0, min(score, 1)), 2)
def reminder_engine(text, friend_list):
    doc = nlp(text)
    if not detect_intent(doc):
        return {"error": "Not a reminder"}
    person = extract_person(doc, friend_list)
    action = extract_action(doc, text)
    location = extract_location(doc, text)
```

```
    if action:
        action = correct_action_text(action)
    confidence = score_confidence(action, location)
    return {
        "intent": "location_based_reminder",
        "action": action,
        "location": location,
        "person": person,
        "confidence": confidence
    }
```

```
import 'dart:async';
import 'package:geolocator/geolocator.dart';
import 'activation_controller.dart';
import 'speed_filter.dart';
import 'topk_manager.dart';
import 'polling_controller.dart';
import 'reminder_engine.dart';
import 'reminder_cache.dart';
import '../services/location_service.dart';
class DynamicDriver{
final SpeedFilter _speedFilter=SpeedFilter();
final ActivationController _activation=ActivationController();
final TopKManager _topK=TopKManager();
final PollingController _polling=PollingController();
```

```
final ReminderEngine _engine=ReminderEngine();
final LocationService _locationService=LocationService();
StreamSubscription<Position>? _positionStream;
Timer? _watcherTimer;
bool _isStarted=false;
List<Map<String,dynamic>> _cachedReminders=[];
Position? _lastPosition;
double _avgSpeed=0;
Future<void> initialize() async{
await ReminderCache.syncFromFirestore();
_cachedReminders=await ReminderCache.loadReminders();
}
void start(){
if(_isStarted)return;
_isStarted=true;
_positionStream=_locationService.getPositionStream().listen(_onLocationUpdate);
}
void stop(){
_positionStream?.cancel();
_watcherTimer?.cancel();
_isStarted=false;}
void _onLocationUpdate(Position position){
double instantSpeed=position.speed*3.6;
_avgSpeed=_speedFilter.addSpeed(instantSpeed);
```

```
_activation.update(_avgSpeed);
if(!_activation.isActive)return;
if(_cachedReminders.isEmpty)return;
if(_lastPosition!=null){
final moved=_engine.haversine(
_lastPosition!.latitude,
_lastPosition!.longitude,
position.latitude,
position.longitude);
if(_topK.currentTopK.isEmpty||_topK.shouldRecompute(moved)){
_topK.computeTopK(position.latitude,position.longitude,_cachedReminders);}
}else{
_topK.computeTopK(position.latitude,position.longitude,_cachedReminders);}
_lastPosition=position;
_scheduleWatcher(position);}
void _scheduleWatcher(Position pos){
_watcherTimer?.cancel();
if(_topK.currentTopK.isEmpty)return;
final r=_topK.currentTopK.first;
final distance=_engine.haversine(
pos.latitude,pos.longitude,r["latitude"],r["longitude"]);
final radius=_engine.dynamicRadius(_avgSpeed);
final remaining=_engine.remainingDistance(distance,radius);
final interval=_polling.computeInterval(remaining,_avgSpeed);
```

```
_watcherTimer=Timer(Duration(seconds:interval),(){
_triggerCheck(r);});}
void _triggerCheck(Map<String,dynamic> r) async{
final pos=await _locationService.getCurrentPosition();
final distance=_engine.haversine(
pos.latitude,pos.longitude,r["latitude"],r["longitude"]);
final radius=_engine.dynamicRadius(_avgSpeed);
if(distance<=radius){
_onTrigger(r);}}
bool _triggered=false
void _onTrigger(Map<String,dynamic> r){
if(_triggered)return;
_triggered=true;
if(onAlarm!=null){
onAlarm!(r);
}
}
void Function(Map<String,dynamic>)? onAlarm;
}
```

A2. SYSTEM SCREENSORTS

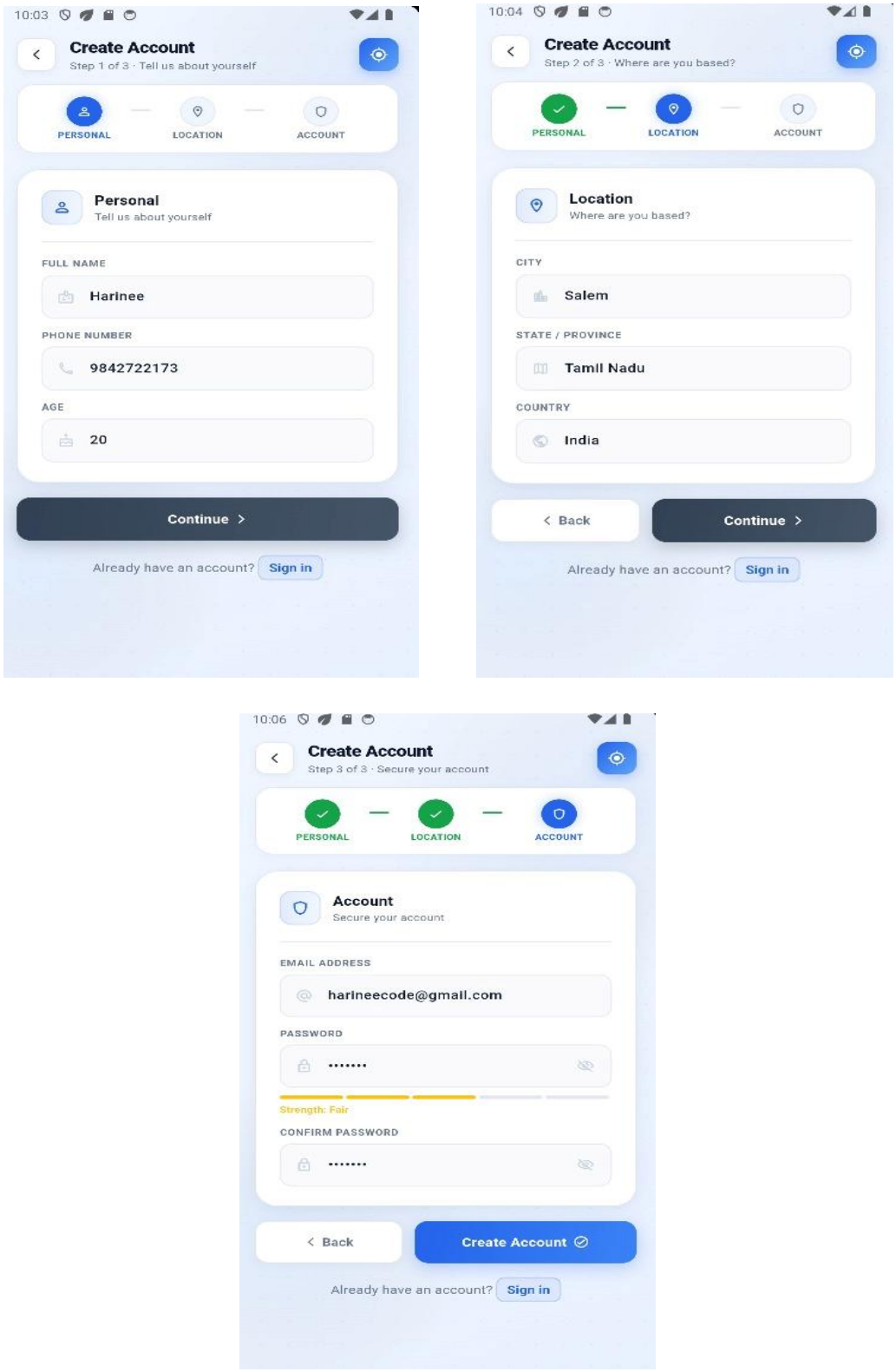


Figure A2.1: Create Account

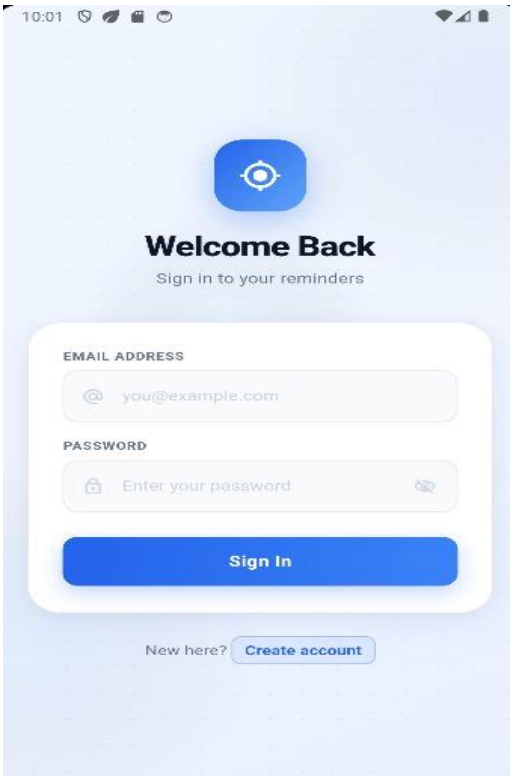


Figure A2.2: Sign in

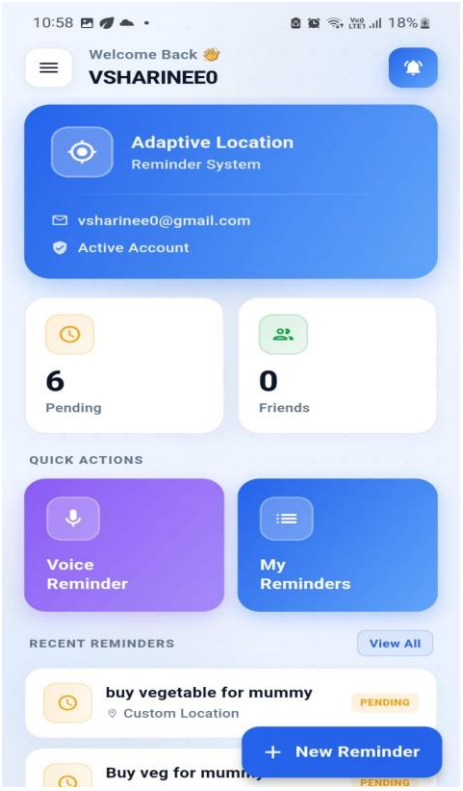


Figure A2.3: Dashboard

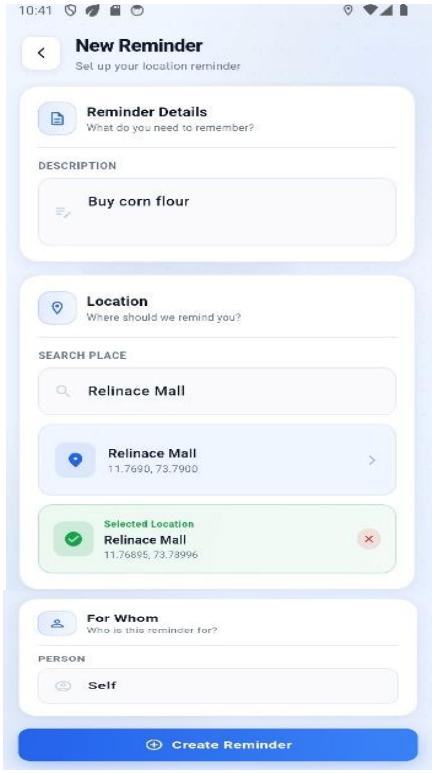


Figure A2.4 : Create reminders



Figure A2.5: Location

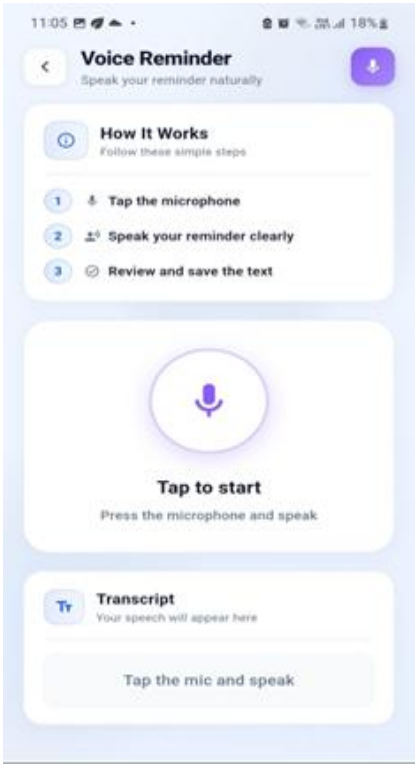


Figure A2.6: Voice Reminder

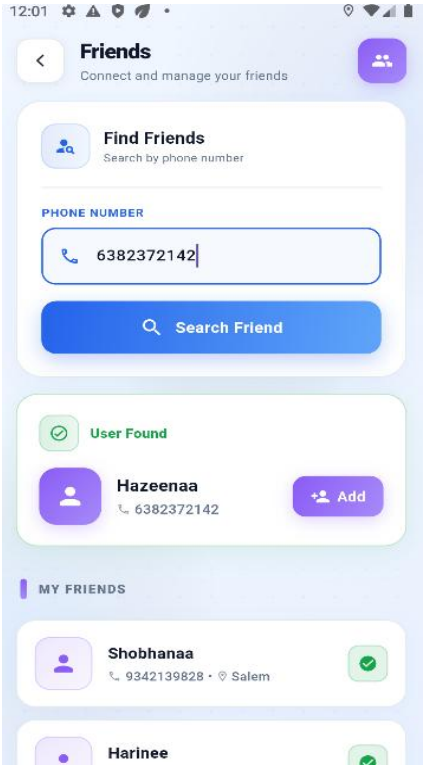


Figure A2.7: Friends

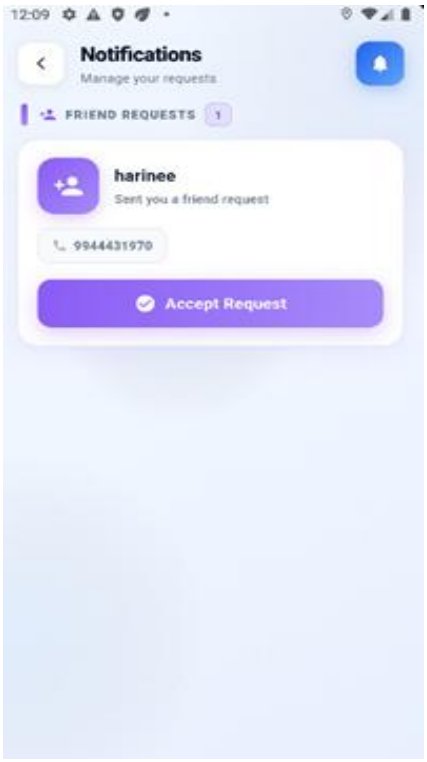


Figure A2.8: Notifications

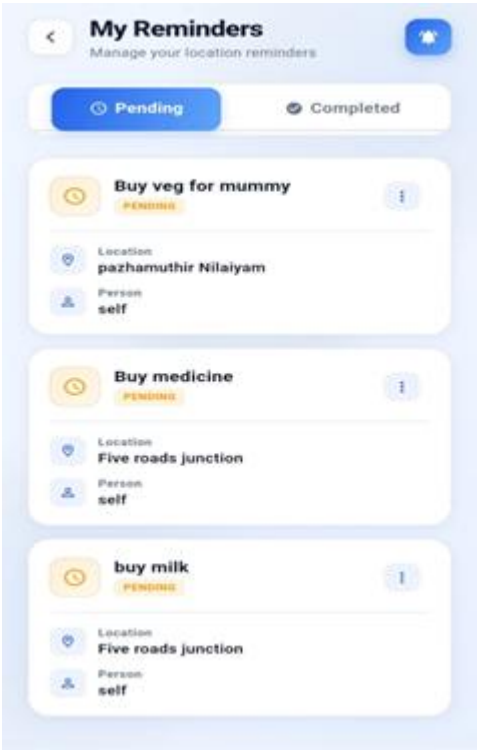


Figure A2.9: Reminders

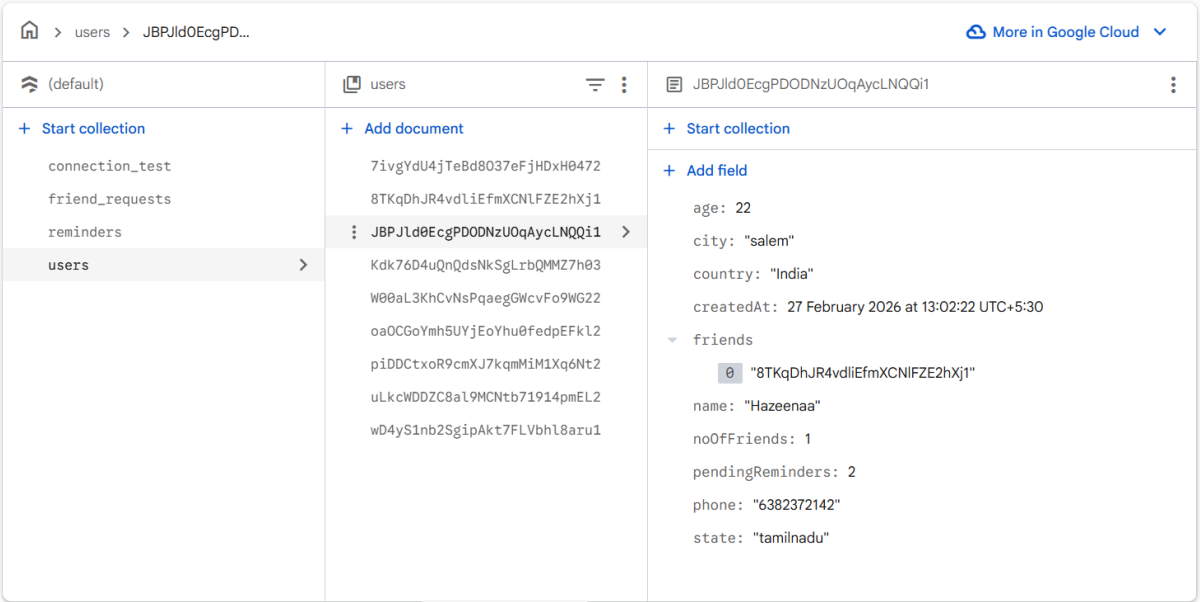


Figure A2.10: Database Users

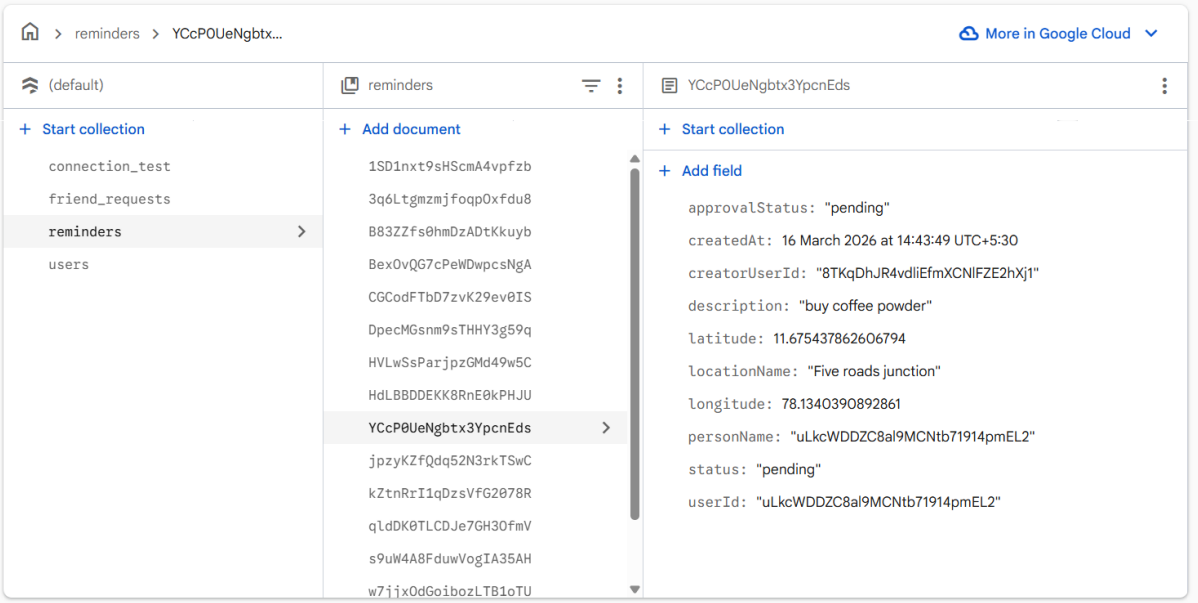


Figure A2.11: Database Reminders

SDG GOALS

SDG 11: Sustainable Cities and Communities

Adaptive Location-Based Reminder System Using Speed-Aware Geofencing strongly aligns with SDG 11, which focuses on making cities and human settlements inclusive, safe, resilient, and sustainable. The system enhances smart urban living by utilizing real-time location tracking and adaptive geofencing to deliver context-aware reminders based on a user's movement and surroundings. By intelligently adjusting reminder triggers using speed-aware algorithms, it ensures timely notifications without unnecessary interruptions, thereby improving efficiency in daily activities such as commuting, task management, and resource utilization.

By reducing missed tasks, unnecessary travel, and cognitive overload, the system contributes to more organized and efficient urban lifestyles. It supports the development of smart city ecosystems by integrating AI-driven decision-making with location-based services, ultimately promoting sustainable living, improved mobility, and better quality of life for individuals in modern urban environments.

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